

STANDARD OPERATING PROCEDURE

Server Whitelisting

Objective

To ensure whitelisting of server is done in correct manner

Participants

1. Technical Lead: Technical Lead (TL) of the respective project.
2. Project Manager(PM): PM of the respective project.
3. MSP Team: PM & Engineer for MSP project.

Understanding the basics

What is Whitelisting?

- Whitelisting, , is a security practice used to control access to a server, network or specific applications by allowing only pre-approved (trusted) IP addresses or IP ranges to interact with the resources (Specific server device,application,or network).
- When a IP address or range IP is whitelisted, it means that the server's configuration or firewall settings have been configured to permit traffic only from this specified IP addresses while blocking all other incoming requests.

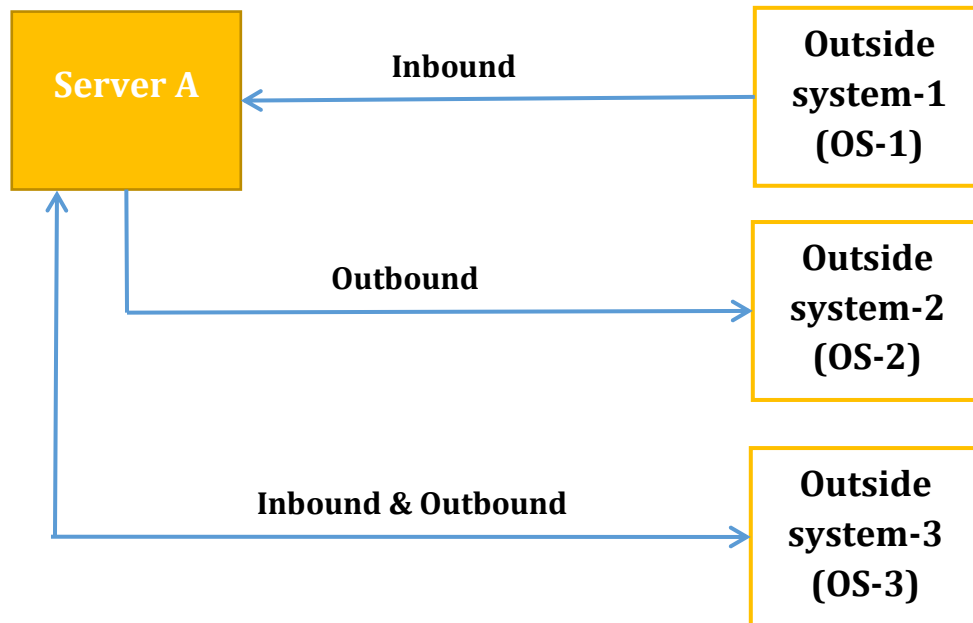
What is Inbound & Outbound?

Inbound:An inbound rule is a configuration setting used in computer networking and firewall management. It specifies the criteria and conditions under which incoming network traffic is allowed to access a specific server device, application, or network.

Inbound rules are typically used to control access to a particular system server from external sources, such as the internet or other networks.

Outbound:An outbound rule, also known as an egress rule, is a network configuration setting used to control and manage the flow of outgoing network traffic from a specific device, application, or network.

Unlike inbound rules that govern incoming traffic, outbound rules focus on regulating the data leaving a system or network and are commonly employed in firewalls and other network security measures.



In Inbound and Outbound Configuration:

Inbound: OS-1 will be allowed to connect & access Server A resources but server is not allowed to access resources of OS-1.

Outbound: Server A is allowed to access content of OS-2 but OS-2 cannot access content of Server A.

What is Public IP & Private IP?

Public IP: is a globally unique address assigned to a device (computer, server, router, etc.) connected to the internet. It acts as an identifier for that device on the public internet and can be accessed from any system/ network connected to internet.

Private IP: is used within a local network, such as a home or office network, to identify devices and facilitate communication between them. It is not directly accessible from the public internet and can only be accessed on local network

STANDARD OPERATING PROCEDURE

Server Whitelisting

Procedure to follow

1. Whenever there is any requirement for whitelisting of any url or IP of server on any of our production servers the respective TL of the project will initiate request by creating ticket in GEL Jira towards issue type **Firewall Whitelist Request** and fill in and attach the form mentioned in Annexure B and also add PM as watcher and mark to CTO for approval (D).
2. If all details are okay, CTO will approve the ticket and mark the same to the PM-MSP team (D1a=D=1). In case the details mentioned are improper, CTO will mark the ticket back to the TL for rectification/clarification (D1b=D+1).
3. Once the request is received by the PM-MSP, MSP team will fill in the Firewall Port opening form and submit it to the respective PM to get the same approved from the respective department head for onward submission to NIC (D2=D1+2).
4. The MSP team will follow and get the signed and stamped copy of the request form from the respective PM within maximum 2 working days (D3=D2+2).
5. Once the signed form is received, MSP team will initiate process as per NIC guidelines and raise request for whitelisting with NIC for servers hosted at NIC cloud (D4=D3+1).
6. The MSP team has to ensure all mail/forms /communications made in terms of this is uploaded in GEL TM under important documents.
7. MSP team should followup with NIC team to get the whitelisting done in a time-bound manner (D5=D4+5).
8. Once the whitelisting is done, the PM-MSP team will mark the ticket back to the TL who initiated the request and update the status of the ticket as Whitelisting Completed.
9. The TL will verify and close the ticket after that.

Logging in JIRA/ TM

Issue Type*: Firewall Whitelist Request
(CTO)

Assignee*: Chief Technical Officer

Reporter (Ticket to be created by)*: Technical Lead (TL)

Description: Request for Whitelisting of <<URL>>

Priority*: Medium

Watcher: Performance Manager (PM)

Original Estimate*: 2 Hrs

Due date*: 7 days

Work logging by*: MSP Team

Status: Close after whitelisting completed by MSP Team.

Project: Project towards which the server belongs.

STANDARD OPERATING PROCEDURE

Server Whitelisting

Requests received from TL on day-D

Seq	Task	Who Does the task	D	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
D	Raise a ticket for submitting Annexure A	TL																
D1a	Approval of ticket from CTO	PM-MSP, TL																
D1b	Marking ticket for rectification/clarification to PM	PM																
D2	Submitting Firewall Port opening form to PM	PM-MSP																
D3	Receiving stamped copy of request form from PM	PM-MSP																
D4	Raise request to NIC for whitelisting	PM-MSP																
D5	Followup with NIC team to get the whitelisting done	PM-MSP, NIC																

STANDARD OPERATING PROCEDURE

Server Whitelisting

ANNEXURE A - SERVER FIREWALL REQUEST FORM

Department Name*: _____ *(Who owns the source server/Where application/database is hosted) Application/server name if any*: _____ Public IP address*(if any) [A]: _____ Private IP address*[B]: _____			
PRIVATE IP rule to access from above [B] server the services of another outside system/Server IP or URL [OUTBOUND]			
Private IP of server which needs to be accessed from [B] server	URL of the server which needs to be accessed from [B] server	Port of the outside server which needs to be accessed from [B] server	Reason
PRIVATE IP rule to access above [B] server services(ports) from identified/other system/servers. [INBOUND]			
Access from Private IP	Access from URL	Port of server[B] which needs to be accessed from other system	Reason
PUBLIC IP rule to access from above [A] server the services of another outside system/Server IP or URL [OUTBOUND]			
Public IP of outside server which needs to be accessed from [A] server	Public URL of the outside server which needs to be accessed from [A] server	Port of the outside server which needs to be accessed from [A] server	Reason

STANDARD OPERATING PROCEDURE

Server Whitelisting

PUBLIC IP rule to access above [A] server services(ports) from identified/other system/servers.
[INBOUND]

Access from Public IP	Access from URL	Port of server[A] which needs to be accessed from other system	Reason

IMPORTANT NOTE:

- The communication between application and database should be compulsorily done on private IP.
- The Private IP rules can only contain the private IP of the servers and cannot be done with public IP. Similarly, the public ip rules can only be done with public IPs or URL and not with the private IPs.
- In case of any confusion, discuss with CTO and raise a request
- Refer Annexure B for sample use cases and example rule implementation

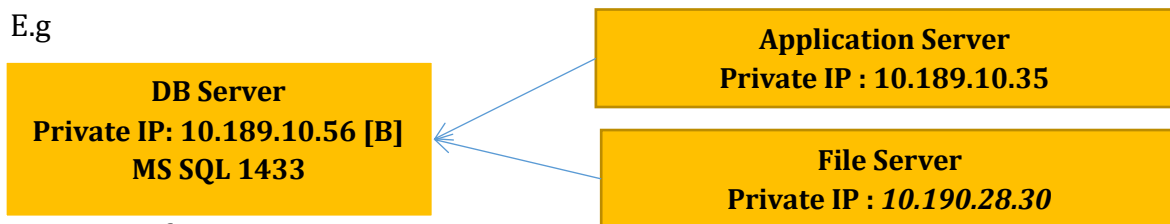
Annexure B Sample use cases for reference

1. Inbound rule : To allow the other / outside system to access the server services (ports)

Use Cases:

- Allow database access to application server on internal (private) IP
- Allow all the systems outside over internet to access the website hosted on public IP
- Allow specific system to access the FTP port of server to transfer data

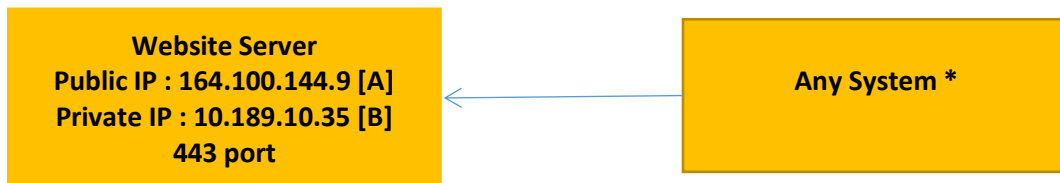
E.g



E.g Request format

PRIVATE IP rule to access above [B] server services(ports) from identified/other system/servers. [INBOUND]			
Access from Private IP	Access from URL	Port of server[B] which needs to be accessed from other system	Reason
10.190.28.30	-	21 (FTP)	Allow FTP access to 10.190.28.30 server.
10.189.10.35	-	1433	Allow db servers port 1433 to be accessed from app server 10.189.10.35

E.g.



PUBLIC IP rule to access above [A] server services(ports) from identified/other system/servers. [INBOUND]			
Access from Public IP	Access from URL	Port of server[A] which needs to be accessed from other system	Reason
E.g. Any IP	-	443	Allow all outside systems to access the site hosted on the A server

STANDARD OPERATING PROCEDURE

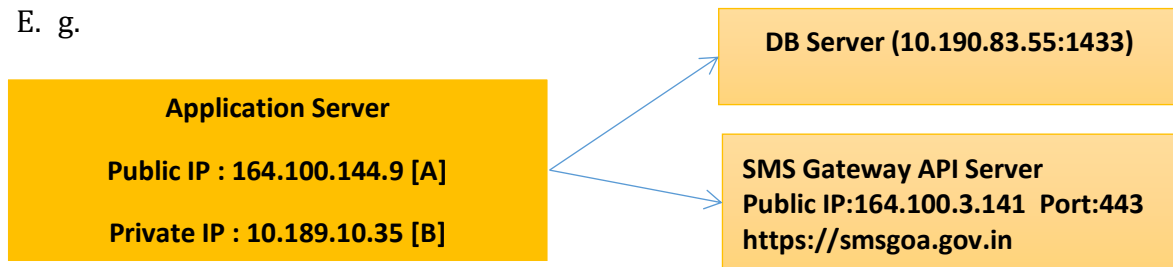
Server Whitelisting

2. **Outbound rule** : Rule to access from current server the services of another outside system/Server IP or URL

Use Cases:

- Connect to the outside database server from current application server on private IP.
- Access the APIs provided by other system from the current application server on 443 port on private or public IP.

E. g.



E.g Request format

PRIVATE IP rule to access from above [B] server the services of another outside system/Server IP or URL [OUTBOUND]			
Private IP of server which needs to be accessed from [B] server	URL of the server which needs to be accessed from [B] server	Port of the outside server which needs to be accessed from [B] server	Reason
E.g. 10.190.83.55	-	1433	Allow app server to access database hosted on 10.190.83.55 on 1433 port

PUBLIC IP rule to access from above [A] server the services of another outside system/Server IP or URL [OUTBOUND]			
Public IP of outside server which needs to be accessed from [B] server	Public URL of the outside server which needs to be accessed from [B] server	Port of the outside server which needs to be accessed from [B] server	Reason
E.g. 164.100.3.141	smsgoa.gov.in	443	Allow current server to access SMS API hosted at smsgoa.gov.in from app server